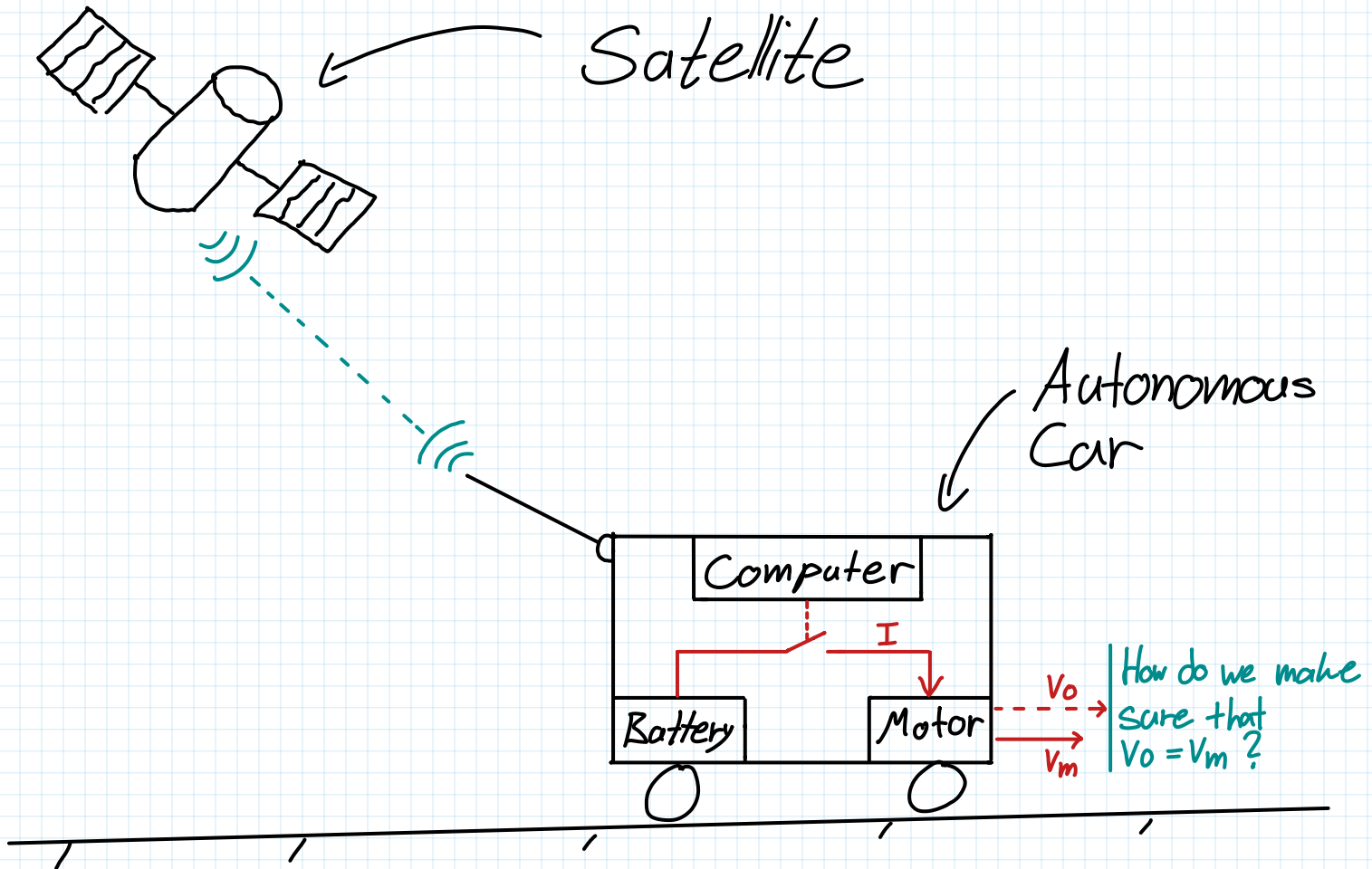


What is control?

- Let's build intuition with an example



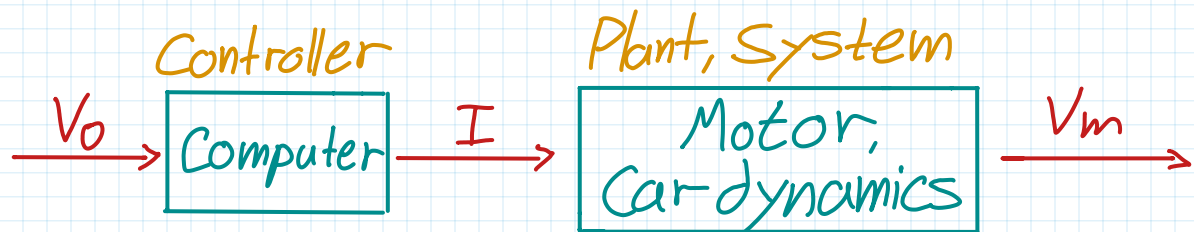
- We command the car's computer to move the car with speed V_o .
- The computer can only control the current I going into the motor.
- The computer knows the car's true speed through measurement V_m of the satellite.

How does the computer know how much current it should send?

⇒ This is the central question of control

- To solve this we will first need to learn many techniques. For you to keep an overview, you must keep the greater picture in mind.

⇒ Let's abstract the car example
(For now without the satellite)



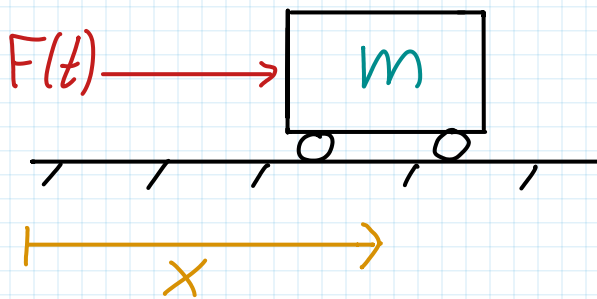
- V_0 , I , V_m are called signals.
- $Computer$, $Motor$ are called systems.

⇒ A system manipulates a signal

What are signals and systems?

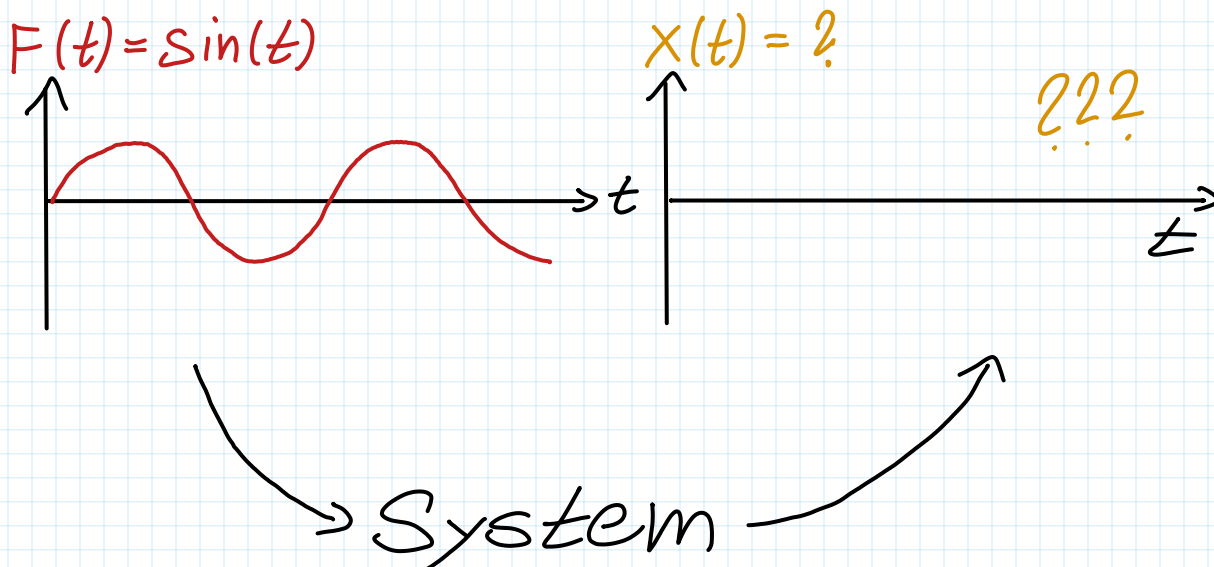
- Signal: A function of time (or frequency)
- System: Any model that possibly has inputs and/or outputs

Example:



- Mass m
- force $F(t)$
- position x

- System: The car with mass m
- Signals: $F(t)$ (input), $x(t)$ (output)



• We can model systems with differential equations

Here:

$$F(t) = m \ddot{X}(t) \Rightarrow \ddot{X}(t) = \frac{1}{m} \cdot \sin(t)$$

$$\Rightarrow \dot{X}(t) = -\frac{1}{m} \cos(t) + C_1$$

$$\Rightarrow X(t) = -\frac{1}{m} \sin(t) + C_1 t + C_2$$

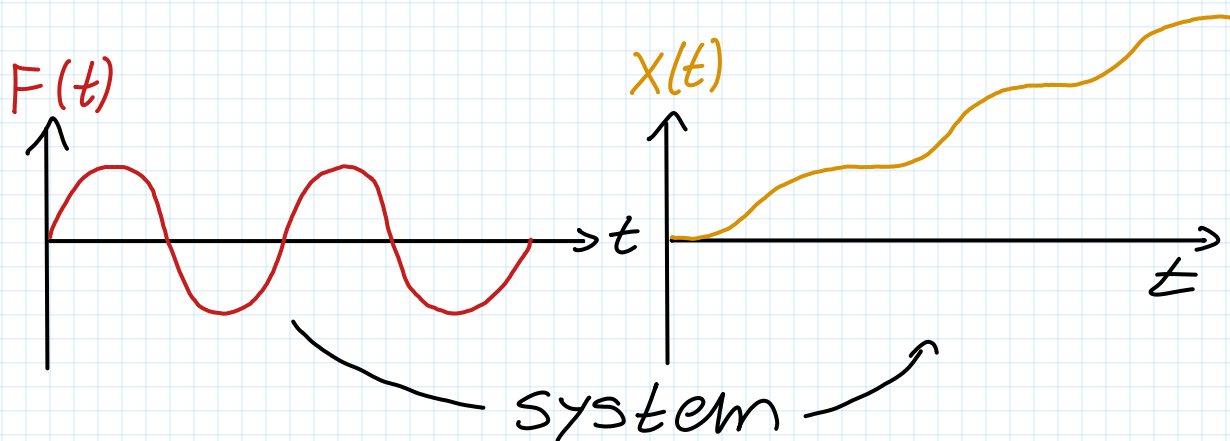
Assume
 $x_0 = 0, v_0 = 0$

From initial conditions: $C_1 = \frac{1}{m}, C_2 = 0$

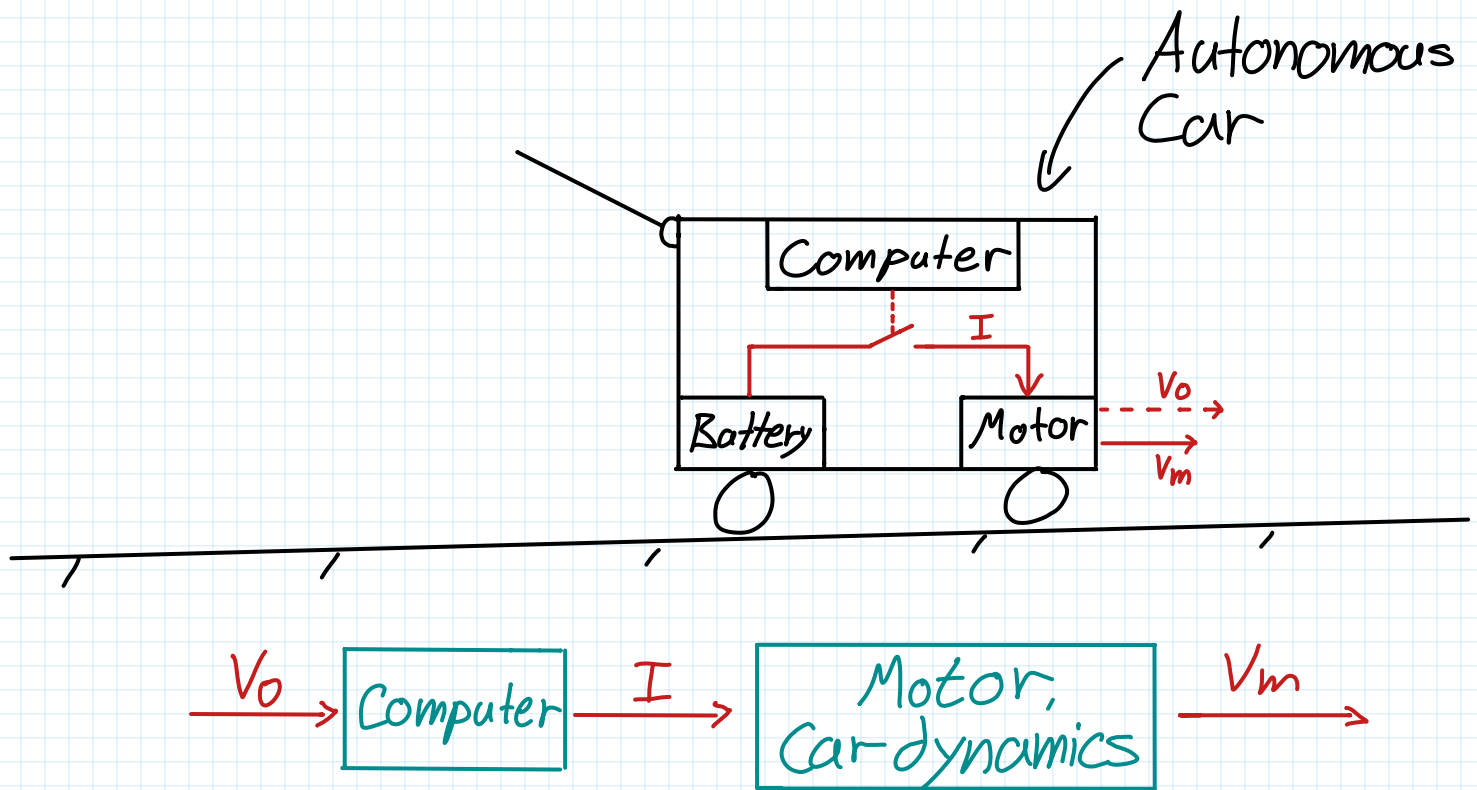
$$X(t) = -\frac{1}{m} \sin(t) + \frac{1}{m} t$$

$$X(t) = \frac{1}{m} (-F(t) + t)$$

$\Rightarrow$ The system is the underlying equation structure and the parameters. It defines how the input signal is manipulated.



Wait, why did we just look at this?



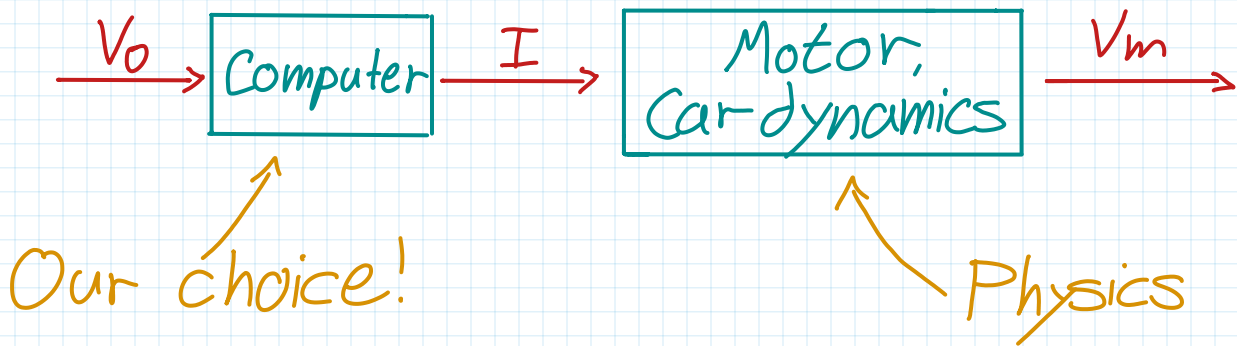
- : Blocks are systems
- →: Arrows are signals

⇒ To figure out how to control the car, we need to know the diff' eqns that describe the behaviour of the blocks!

- This is why the first part of the course will be modeling.

If every system is just defined by physics, then what even is our job as engineers?

⇒ This is not true!!



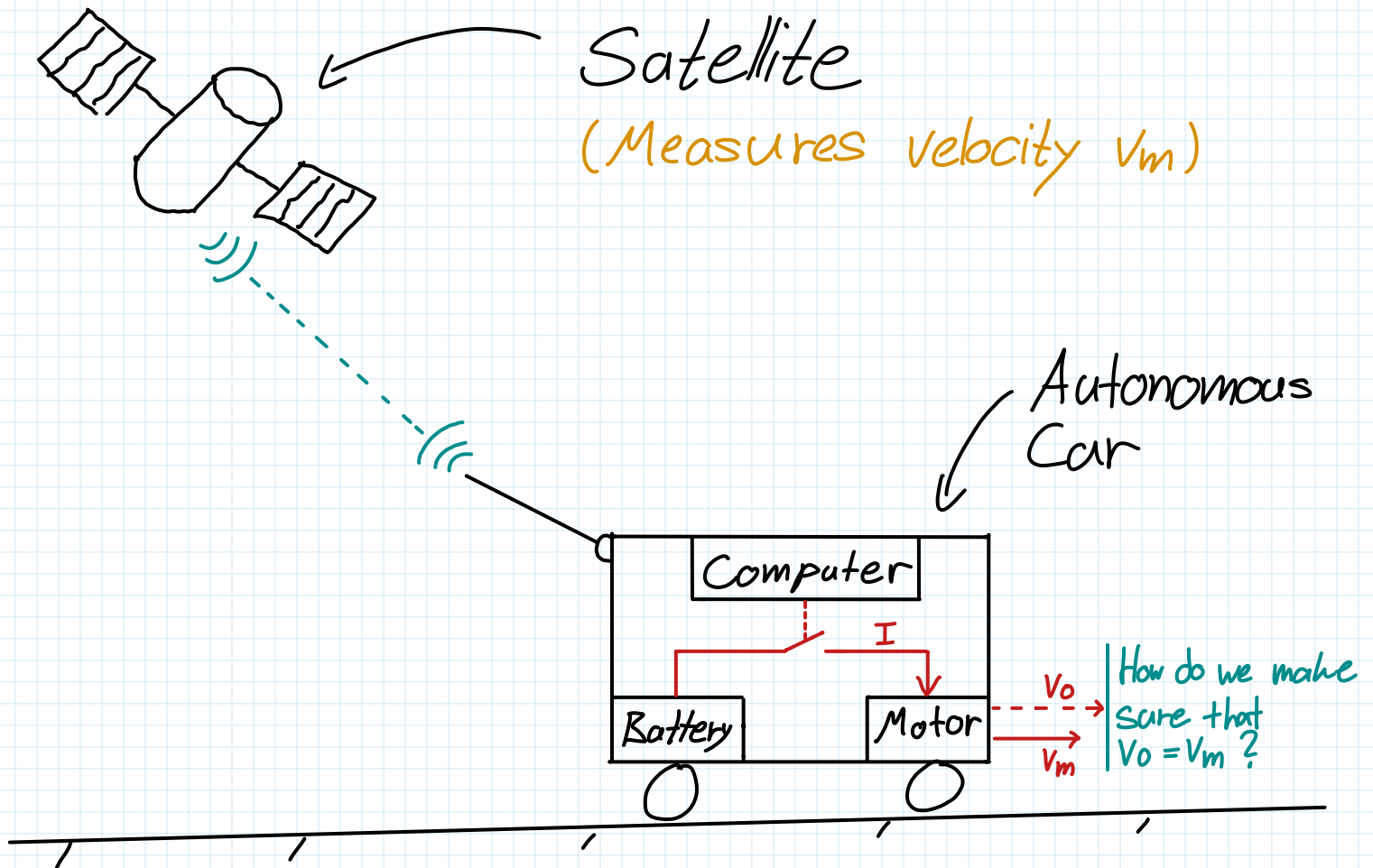
• Some blocks are predefined. The ones that we can design (*computer*), are called Controllers.

⇒ We need to design controllers such that the total system does what we want. This is hard because of:

- Model uncertainties
- External disturbances
- Inaccurate sensors
- ...

• These problems will be addressed by introducing Feedback.

⇒ Back to our example



Think about how the satellite could improve this problem...

• All of what I showed you was massively simplified and not covered yet.
(you are not expected to understand it now)

⇒ This was just so you have the big picture in mind!

Intermediate Quiz

Q1) The Input is what we tell the computer to do.

☐ True ☒ False

⇒ We give the computer the reference.
It then calculates what input the motor should receive.

Q2) Each block represents a physical system. ☐ True ☒ False

⇒ The controller is a software program.

Q3) Not all control systems are electrical.

☒ True ☐ False

⇒ Example: Watt governor